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St. Joseph's College of Arts & Science (Autonomous)

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B. Sc DEGREE EXAMINATION-APRIL 2021

BC406S-ANALYTICAL BIOCHEMISTRY-II

Time: 3 hrs **Max Marks: 75**

SECTION-A (25X1=25)

Answer ALL the questions:

I.Choose the best answer (10X1=10)

1. Chromatography is used to separate

   a) Solution b) Mixture c) Molecules d) Atoms

2. Thin layer chromatography is

   a) partition chromatography b) Electrical mobility of ionic species

   c) Adsorption chromatography d) based on molecular weight

3. Locating agent of amino acids is

   a) Diazo reagent b) Ninhydrin spray

   c) Ampoteric oxides d) Neutral oxides

4. Silica gel-G is used in_____

   a) HPLC b) TLC c) GLC d) Column chromatography

5. The technique electrophoresis was developed by

   a) Sanger b) Tswett c) Tiselius d) Svedberg

6. The most common type of gel used in DNA separation is

   a) Agar b) Agarose c) Polyacrylamide d) Cellulose

7. Western blotting is a technique to measure _____

   a) DNA b) RNA c) Protein d) Lipid

8. Half life of C-14 is

   a) 10 years b) 3587 years c) 30 days d) 5732 years.

9. Nuclei with atomic number 1 to 82 are

   a) Stable b) Unstable c) Radioactive d) Moderately stable

10. Geiger Muller counter cannot detect

    a) Beta rays b) X rays c) Gamma rays d) Cosmic rays

II. Say true or false (5X1=5)

11. Rf value of any solute is always below zero
12. TLC can be used for small scale preparative work
13. SDS is a cationic detergent
14. Cosmic rays are ionization radiation.
15. Unit of radio activity is Curie.

III. Answer in one sentence (10X1=10)

16. Define chromatography.
17. Mention the phases of chromatography.
18. Define retention time**.**
19. What is a molecular sieve?
20. Give the importance of stacking gel
21. Give the principle of northern blotting
22. Give the components of atom.
23. Define Radioactive decay
24. What do you meant by half life?
25. What is radiodating?

SECTION-B (4X5=20)

Answer any FOUR of the following:

26. Write down the principle and application of paper chromatography.
27. Write a short note on reverse phase chromatography.
28. Give a brief note on Southern blotting technique..
29. Give an account of Tiselius moving boundary electrophoresis.
30. Discuss the principle and application of GM counter.
31. Comment on Biological hazards of radiation and its safety aspects.

SECTION-C (3X10=30)

Answer any THREE of the following:

32. Explain in detail the principle, operation and applications of GLC.
33. Describe the procedure for ion exchange chromatography.
34. Write an essay on SDS-PAGE.
35. Explain the principle, procedure and applications of RIA.
36. Elaborate on the application of radio isotopes in the field of biology.

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